📚 Library ERP – Complete Installation & Deployment Guide

Step-by-step instructions for **Windows** (XAMPP / WAMP), **macOS** (MAMP / native), and **Linux** (Ubuntu / CentOS). Includes **performance tuning**, **security hardening**, and **error-free** deployment practices.

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1. 📥 Download from GitHub

📥 Download from GitHub

Repository: **sameemonm1-hue/Library-Management-System**

⬇ Download ZIP

★ Star the repository on GitHub to support the project!

2. ⚙️ System Requirements

Component	Minimum Requirement
Web Server	Apache 2.4+ (or Nginx with PHP-FPM)
PHP	**8.1**, 8.2 or **8.3** (recommended)
PHP Extensions (critical)	pdo_sqlite, sqlite3, zip, gd, fileinfo, mbstring, curl, json, xml, intl
Memory	256 MB (512 MB recommended)
Disk Space	1 GB + space for uploads

> 💡 **Performance tip:** Enable OPCache for faster execution. In `php.ini`, set `opcache.enable=1` and `opcache.memory_consumption=128`.

3. 🪟 Windows – XAMPP

3.1 Install XAMPP

- Download XAMPP for Windows from **apachefriends.org** (PHP 8.2 or 8.3 version).
- Run the installer – accept defaults (`C:\xampp`).
- Launch XAMPP Control Panel and start **Apache**.

3.2 Enable Required PHP Extensions

- Open `C:\xampp\php\php.ini` in Notepad.
- Uncomment (remove the leading `;`) these lines:

```
extension=sqlite3
extension=gd
extension=zip
extension=curl
extension=mbstring
extension=fileinfo
extension=intl
```

- **Save** and **restart Apache** from the XAMPP Control Panel.

3.3 Deploy the Application

- Extract the downloaded `Library-Management-System-main.zip`.

- Inside it, find and extract `library-system.zip` → you get the `library-system` folder.
- Copy the `library-system` folder to `C:\xampp\htdocs\` .

3.4 Set Permissions

Right-click the `uploads` , `backups` , and `config` folders inside `library-system` → Properties → Security → give **Modify/Write** permission to `Everyone` or the web server user.

Alternatively, run as Administrator:

```
icacls "C:\xampp\htdocs\library-system\uploads" /grant Everyone:F /T
icacls "C:\xampp\htdocs\library-system\backups" /grant Everyone:F /T
icacls "C:\xampp\htdocs\library-system\config" /grant Everyone:F /T
```

3.5 Access the System

Open browser: http://localhost/library-system/

- **Default login:** `admin` / `admin123`
- **OPAC:** http://localhost/library-system/opac
- **Visitor Log:** http://localhost/library-system/visitor

⚙ RESET TO DEFAULT – System Configuration

After installation, you can restore all system settings to their **factory defaults** with a single click.

Navigation **Admin Dashboard → System Settings → System Configuration**

Action Click the **RESET TO DEFAULT** button – all settings will be restored to their original values.

> 💡 This is useful if you have made many changes and want to start fresh, or if you encounter configuration issues.

3.6 Windows – WAMP (Alternative)

- Install WAMP from https://www.wampserver.com/.
- Enable PHP extensions via the WAMP tray icon → PHP → Extensions (enable **php_zip**).
- Place the `library-system` folder inside `C:\wamp64\www\` .
- Access at http://localhost/library-system/

4. 🍎 macOS

4.1 Using MAMP (Recommended)

- Download MAMP from **mamp.info** (free version).
- Install and launch MAMP. Set PHP version to 8.x in preferences.
- Extract `Library-Management-System-main.zip` → extract `library-system.zip` → get the `library-system` folder.
- Copy `library-system` to `/Applications/MAMP/htdocs/` .
- Set permissions (Terminal):

```
cd /Applications/MAMP/htdocs
chmod -R 755 library-system/
chmod -R 777 library-system/uploads library-system/backups library-system/config
```

- **Enable ZipArchive:** Open `/Applications/MAMP/bin/php/php8.x/conf/php.ini` and uncomment `extension=zip` .
- Start MAMP servers and access http://localhost:8888/library-system/

⚙️ RESET TO DEFAULT – System Configuration

After installation, you can restore all system settings to their **factory defaults** with a single click.

Navigation **Admin Dashboard → System Settings → System Configuration**

Action Click the **RESET TO DEFAULT** button – all settings will be restored to their original values.

> 💡 This is useful if you have made many changes and want to start fresh, or if you encounter configuration issues.

4.2 Using Native Apache & PHP (Advanced)

- Install Homebrew: `/bin/bash -c "$(curl -fsSL https://raw.githubusercontent.com/Homebrew/install/HEAD/install.sh)"`
- Install PHP 8.3: `brew install php@8.3`
- Enable ZipArchive: ensure it's enabled in `/usr/local/etc/php/8.3/php.ini` .
- Place `library-system` in `/Library/WebServer/Documents/` :

```
sudo cp -r library-system /Library/WebServer/Documents/
```

- Set permissions:

```
cd /Library/WebServer/Documents
sudo chown -R _www:_www library-system/
sudo chmod -R 755 library-system/
sudo chmod -R 777 library-system/uploads library-system/backups library-system/config
```

- Edit `/etc/apache2/httpd.conf` to enable `mod_rewrite` and set `AllowOverride All`.
- Restart Apache: `sudo apachectl restart`
- Access at http://localhost/library-system/

5. 🐧 Ubuntu / Linux – Complete Setup

📋 Manual Installation – Ubuntu / Linux

Follow these simple steps to download, extract, and deploy the application on your Ubuntu/Linux server.

📥 Step 1: Download from GitHub

Visit the official GitHub repository and download the ZIP file:

- Go to: **https://github.com/sameemonm1-hue/Library-Management-System**
- Click the green **"Code"** button → **"Download ZIP"**
- Or use the **⬇️ Download ZIP** button in the download box at the top of this guide.

> 💡 The downloaded file will be named `Library-Management-System-main.zip`.

📂 Step 2: Extract the Main Archive

Extract the downloaded ZIP file. On Ubuntu, you can right-click the file and select **"Extract Here"** or use the Archive Manager.

> 📂 After extraction, you'll see a folder named `Library-Management-System-main`.

🔍 Step 3: Locate the `library-system.zip` File

Inside the extracted folder, find the **ZIP file** named `library-system.zip` – this is the actual application archive.

```
📂 Folder Structure:
Library-Management-System-main/
  ├── library-system.zip   ← THIS is the application archive
  ├── Installation Guide.html
  └── (other files)
```

> ⚠️ **Important:** Do NOT copy the entire `Library-Management-System-main` folder to your web root. You need to extract `library-system.zip` first.

📦 Step 4: Extract `library-system.zip`

Right-click `library-system.zip` and select **"Extract Here"** to get the actual application folder.

> ✅ After extraction, you now have the `library-system` folder – this is the application folder.

📂 Step 5: Place the Application in Root (Home)

Move or copy the `library-system` folder to your home directory for easy access.

> 💡 You can simply drag and drop the `library-system` folder to your home directory.

💻 Step 6: Open Terminal & Switch to Root

Open a terminal by pressing **Ctrl + Alt + T** or searching for "Terminal" in the applications menu.

Type the following command to switch to the root user:

```
$ sudo su
```

Enter your password when prompted. The terminal prompt will change to `root@home:~#` indicating you are now logged in as root.

> ⚠️ **Security Note:** Use `sudo su` with caution. Always type `exit` to leave root mode after completing the deployment.

📋 Step 7: Copy the Application to Web Root

As root, copy the `library-system` folder from your home directory to Apache's web root:

```
root@home:~# sudo cp -r ~library-system /var/www/html/
```

> ✅ The `library-system` folder is now deployed at `/var/www/html/library-system/`

🔧 Step 8: Set Correct Permissions

Set ownership and permissions so Apache can read and write to the application. Run these commands in the terminal:

```
root@home:~# cd /var/www/html
root@home:/var/www/html# sudo chown -R www-data:www-data
/var/www/html/library-system/
root@home:/var/www/html# sudo chmod -R 755 /var/www/html/library-system/
root@home:/var/www/html# sudo chmod -R 777 /var/www/html/library-
system/uploads/
root@home:/var/www/html# sudo chmod -R 777 /var/www/html/library-
system/backups/
root@home:/var/www/html# sudo chmod -R 777 /var/www/html/library-
system/config/
root@home:/var/www/html# exit
```

> 💡 The `config/` folder needs write permission so the system can create the SQLite database file (`library.db`).

🎯 Step 9: Access the System

Open your browser and navigate to the system URL:

- **Main System:** `http://localhost/library-system/` (or `http://your-server-ip/library-system/`)
- **OPAC:** `http://localhost/library-system/opac`
- **Visitor Log:** `http://localhost/library-system/visitor`

Default Login:

- **Username:** `admin`
- **Password:** `admin123`

> ⚠️ **Security Note:** Change the default admin password immediately after first login!

> 💡 **Pro Tip:** Using `git clone` instead of downloading ZIP lets you pull future updates with a single command.

✅ After following these steps, continue with the **Virtual Host** and **Database Creation** sections below.

5.1 Install Apache & PHP 8.3

Run these commands in the terminal to install Apache and PHP:

```
# Update system
$ sudo apt update && sudo apt upgrade -y

# Install Apache, Git, Unzip
$ sudo apt install -y apache2 git unzip

# Install PHP 8.3 and required extensions
$ sudo apt install -y php8.3 libapache2-mod-php8.3 php8.3-sqlite3 \
  php8.3-gd php8.3-zip php8.3-mbstring php8.3-curl php8.3-xml php8.3-intl

# Enable Apache mod_rewrite
$ sudo a2enmod rewrite
$ sudo systemctl restart apache2
```

5.2 Configure Virtual Host (Recommended)

```
# Create a new virtual host configuration
$ sudo nano /etc/apache2/sites-available/library.conf
```

Paste the following:

```
<VirtualHost *:80>
    ServerName library.local
    DocumentRoot /var/www/html/library-system
    <Directory /var/www/html/library-system>
        Options Indexes FollowSymLinks
        AllowOverride All
        Require all granted
    </Directory>
    ErrorLog ${APACHE_LOG_DIR}/library-error.log
    CustomLog ${APACHE_LOG_DIR}/library-access.log combined
</VirtualHost>
```

```
# Enable the site and reload Apache
$ sudo a2ensite library.conf
$ sudo a2dissite 000-default.conf
$ sudo systemctl reload apache2

# Add to /etc/hosts for local access
$ echo "127.0.0.1 library.local" | sudo tee -a /etc/hosts
```

5.3 Database Creation – Important!

⚠️ REFRESH – repeat three or more times for creating DB

If the database is **not** created automatically on the first visit, simply **refresh the page 3–4 times**. The system may need a few requests to initialise all tables.

If the database still doesn't appear, ensure the `config/` folder is writable (already set in permissions above). Additionally, run these commands to set full permissions on the `uploads/` and `backups/` folders:

```
$ cd /var/www/html
$ sudo chmod -R 777 /var/www/html/library-system/uploads/
$ sudo chmod -R 777 /var/www/html/library-system/backups/
```

✅ After running the commands, refresh the page again – the database should be created.

⚙️ RESET TO DEFAULT – System Configuration

After installation, you can restore all system settings to their **factory defaults** with a single click.

Navigation **Admin Dashboard → System Settings → System Configuration**

Action Click the **RESET TO DEFAULT** button – all settings will be restored to their original values.

💡 This is useful if you have made many changes and want to start fresh, or if you encounter configuration issues.

6. 🐧 CentOS / RHEL / Fedora

6.1 Install Dependencies

```
# Install EPEL and Remi repositories
$ sudo dnf install epel-release
$ sudo dnf install https://rpms.remirepo.net/enterprise/remi-release-9.rpm
$ sudo dnf module reset php
$ sudo dnf module install php:remi-8.3

# Install Apache and PHP extensions (including zip)
$ sudo dnf install httpd php php-sqlite3 php-gd php-zip php-mbstring \
  php-curl php-xml php-intl git unzip

# Enable and start Apache
$ sudo systemctl enable httpd
$ sudo systemctl start httpd
```

6.2 Deploy the Application

Follow the **Manual Installation Steps** from the Ubuntu section above (Steps 1-8).

```
# After extraction, copy to web root
$ sudo cp -r ~/library-system /var/www/html/

# Set permissions for CentOS/RHEL
$ sudo chown -R apache:apache /var/www/html/library-system/
$ sudo chmod -R 755 /var/www/html/library-system/
$ sudo chmod -R 777 /var/www/html/library-system/{uploads,backups,config}

# Allow .htaccess overrides
$ sudo nano /etc/httpd/conf/httpd.conf    # Set AllowOverride All

# Restart Apache
$ sudo systemctl restart httpd
```

Access at `http://your-server-ip/library-system/`

⚙️ RESET TO DEFAULT – System Configuration

After installation, you can restore all system settings to their **factory defaults** with a single click.

📍 Navigation **Admin Dashboard → System Settings → System Configuration**

🔘 Action Click the **RESET TO DEFAULT** button – all settings will be restored to their original values.

> 💡 This is useful if you have made many changes and want to start fresh, or if you encounter configuration issues.

7. ⚡ Performance Tuning

7.1 PHP Settings (Recommended)

Edit your `php.ini` file to match these values for optimal performance:

```
memory_limit = 512M
upload_max_filesize = 100M
post_max_size = 100M
max_execution_time = 600
max_input_time = 600
date.timezone = Asia/Kolkata

# Enable OPCache for faster execution
opcache.enable = 1
opcache.memory_consumption = 128
opcache.max_accelerated_files = 10000

# Enable error logging
error_reporting = E_ALL & ~E_DEPRECATED & ~E_STRICT
display_errors = Off
log_errors = On
error_log = /var/log/php_errors.log
```

After changes, restart your web server.

7.2 Enable SQLite WAL Mode

The system automatically enables `PRAGMA journal_mode=WAL` for better concurrency and performance – no action required.

7.3 Configure .htaccess for Security

Ensure the following `.htaccess` file is present in the root of your `library-system` folder:

```
<IfModule mod_rewrite.c>
    RewriteEngine On
    RewriteRule ^(config|backups|vendor|logs|temp|cache)/ - [F,L]
    Options -Indexes
</IfModule>
```

8. 📊 Composer for Excel & PDF Reports

If you need advanced reporting (Excel, PDF), install Composer and the required packages:

```
$ cd /var/www/html/library-system
$ sudo apt install -y composer
$ composer require phpoffice/phpspreadsheet
$ composer require dompdf/dompdf
```

> If Composer is not available, the system falls back to HTML/print-based PDFs.

9. ⏰ Scheduled Tasks (Cron)

Set up these cron jobs to automate overdue notifications, email processing, and backup rotation:

```
$ sudo crontab -e
```

```
# Daily tasks (overdue reminders, fine calculation)
0 2 * * * /usr/bin/php /var/www/html/library-system/cron/daily_tasks.php >>
/var/log/library-cron.log 2>&1

# Process email queue every 5 minutes
*/5 * * * * /usr/bin/php /var/www/html/library-system/cron/process_email_queue.php >>
/var/log/library-email.log 2>&1

# Weekly backup rotation
0 3 * * 1 /usr/bin/php /var/www/html/library-system/cron/weekly_tasks.php >>
/var/log/library-cron.log 2>&1
```

> Adjust paths if your installation is in a different directory.

10. 🔒 Security Hardening

- **Change default passwords** immediately after login (admin, staff1).
- **Enable HTTPS** using Let's Encrypt: `sudo certbot --apache -d yourdomain.com`
- **Restrict access** to `config/`, `backups/`, `uploads/` via `.htaccess`.
- **Enable fail2ban** to block brute-force attacks on login.
- **Regular updates**: `sudo apt update && sudo apt upgrade -y`.
- **Pull latest code** from GitHub periodically.

11. 💾 Backup & Restore

11.1 Create a Backup

- Log in as admin → **System Settings** → **Backup**.
- Click **"Create Backup"** – a `.zip` file will be saved in the `backups/` folder.
- Backups include the SQLite database and all uploaded files.

11.2 Restore from a Backup

- In the same Backup section, select a previous backup and click **"Restore"**.
- The system will replace the current database and files with the backup version.

> ⚠ **Always create a fresh backup before performing a restore or update.**

12. 🔧 Troubleshooting Quick Reference

Issue	Solution
ZipArchive not available	Enable `extension=zip` in php.ini and restart the web server.
White screen / 500 error	Check Apache error logs: `sudo tail -f /var/log/apache2/error.log`.
Database not created	Refresh the page 3–4 times; ensure `config/` is writable.
Upload fails	Check `uploads/` permissions and PHP `upload_max_filesize`.
Email not sending	Test SMTP in Settings → Notifications.
PDF export blank	Install dompdf via Composer, or use browser print.
Slow performance	Enable OPCache, increase memory limit, ensure SQLite WAL mode is active.

13. 🗑️ Uninstall

```
# For Apache
$ sudo a2dissite library.conf
$ sudo rm -rf /var/www/html/library-system
$ sudo systemctl reload apache2
```

On Windows, simply delete the `library-system` folder from `htdocs`.

14. ⚙️ Reset to Default

⚙️ RESET TO DEFAULT – System Configuration

After installation, you can restore all system settings to their **factory defaults** with a single click.

Navigation **Admin Dashboard → System Settings → System Configuration**

Action Click the **RESET TO DEFAULT** button – all settings will be restored to their original values.

> 💡 This is useful if you have made many changes and want to start fresh, or if you encounter configuration issues.

📋 What Gets Reset?

- **Library Settings:** Fine amount, loan duration, max books per user
- **Email Configuration:** SMTP settings, email templates
- **Notification Settings:** Overdue reminders, due date alerts
- **Security Settings:** Password policies, session timeout
- **Appearance:** Theme, logo, layout settings
- **All other system configurations** to their original values

> ⚠️ **Note:** This does **NOT** delete your database data (books, users, transactions, etc.). Only system **configuration settings** are reset.